STAT4C03/6C03 Generalized Linear Models

Assignment 5 Due at 10pm on Tuesday April 7, 2026

Instructions:

1. The only way to submit your assignment is through the Crowdmark page for the course. Use the link on Avenue (Content > Crowdmark) to login to Crowdmark and follow the instructions there to upload your assignment.

2. It is your responsibility to ensure that your submitted solutions are readable and complete. Any submitted file which cannot be opened will be given a mark of 0.

3. You may use R Markdown or Knitr to compile the R questions in your assignment but are not required to do so. For any question using R, you are required to include both the R code and output in your solutions as well as your written comments on the model(s) fitted.

4. Late assignments will be automatically assessed a penalty of 2% per hour, or part thereof, that they are late according to the Crowdmark time stamp.

5. You may update your submission up until the deadline. Once the due date has passed, you will no longer be able to update any submission. This includes adding questions that you did not submit before the deadline.

6. Please indicate **clearly** on your solutions whether you are in STATS 4C03 or STATS 6C03.

7. You are expected to show all details of your solution and any results taken from my notes or the textbooks must be clearly and properly referenced.

8. Students with accommodations through Student Accessibility Services who require accommodations for this assignment must contact the professor **at least 48 hours before the assignment is due** to request any such accommodations.

9. No other extensions to the due date and time will be given except in extreme circumstances.

10. There are 3 questions with a total of 100 marks available for all students (no extra STATS 6C03 question this time) but the questions are not equally weighted.

Q. 1 **[40 marks]**

The dataset in `Happiness.csv` (`happy` in the `faraway` package) contains data from 39 students in a University of Chicago MBA program. The variables are described on Avenue.

a) The response variable is in the column `happy`. The other variables are the covariates. Convert the covariates sex, love and work into factors and construct plots of the relationship of each of the 4 covariates with the response. Do any of the covariates look like they are not related to the response?

b) Convert the `happy` variable to an ordered factor and fit a proportional odds logistic regression model with main this ordered factor as response and main effects for each of the four covariates. Why are there only 8 estimated intercepts? Construct an index plot of the estimated intercepts. Comment on any deviations from linearity and what this may signify.

c) The `predict` function for `polr` objects can either predict the most likely class for a new individual or the distribution using `"probs"`. Predict the happiness distribution for an individual whose parents make $30,000 per year, who is lonely, not sexually active and has no job. Repeat this for the same individual except now he has a deep feeling of belonging and caring and comment on the two distributions.

d) Test each covariate for significance. If any of the covariates are not significant drop the covariate with the highest p-value. Repeat this process of *backward elimination* to find a model in which all of the covariates are significant. Compare this model to the initial model to ensure that the fitted model is no worse than the model with all 4 covariates.

e) In the models above we considered love and work as factors. Test if these variables can be considered numerical if they appear in the final model from the backwards elimination done in part (d). What is the interpretation of the conclusion of this test? What model would you select for this dataset?

f) Repeat the predictions from part (c) for the final model you would select for this dataset. How have the predicted distributions changed?

Q. 2 **[30 marks]**

The `glm` function in `R` allows for the fitting of a generalized linear model in which the conditional distribution of the response given the covariates has an Inverse Gaussian (or Wald) distribution. This is a distribution on the positive real line with probability density function

$$f(y;\mu,\phi) \;=\; \sqrt{\frac{1}{2\phi\pi y^3}}\exp\left\{-\frac{(y-\mu)^2}{2\phi\mu^2 y}\right\} \qquad \text{for } y>0$$

where $\mu > 0$ and $\phi > 0$ are the parameters of the model.

a) Show that this model can be written in exponential family form identifying the canonical parameter θ and the functions $a(\phi)$, $b(\theta)$, $c(y,\phi)$.

b) Use the results in my notes to show that μ is the mean of this distribution and find the variance.

c) Suppose that there is a single continuous covariate x and we use the canonical link function,

$$\frac{1}{\mu^2} \;=\; \beta_0 + \beta_1 x$$

Give the log-likelihood for $l(\boldsymbol{\beta})$ and the score vector $\boldsymbol{U}(\boldsymbol{\beta})$ for $\boldsymbol{\beta}$ where

$$\boldsymbol{U}(\boldsymbol{\beta}) \;=\; \begin{pmatrix} U_0(\boldsymbol{\beta}) \\ U_1(\boldsymbol{\beta}) \end{pmatrix} \;=\; \begin{pmatrix} \frac{\partial l(\boldsymbol{\beta})}{\partial \beta_0} \\ \frac{\partial l(\boldsymbol{\beta})}{\partial \beta_1} \end{pmatrix}$$

for a fixed dispersion parameter ϕ.

d) Use the results from part (b) to show that $\mathrm{E}\left(U_0(\boldsymbol{\beta})\right) \;=\; \mathrm{E}\left(U_1(\boldsymbol{\beta})\right) \;=\; 0$.

e) Use the results from part (b) to find the covariance matrix of the score

$$\mathrm{Var}\left(\boldsymbol{U}(\boldsymbol{\beta})\right) \;=\; \begin{pmatrix} \mathrm{Var}\left(U_0(\boldsymbol{\beta})\right) & \mathrm{Cov}\left(U_0(\boldsymbol{\beta}), U_1(\boldsymbol{\beta})\right) \\ \mathrm{Cov}\left(U_0(\boldsymbol{\beta}), U_1(\boldsymbol{\beta})\right) & \mathrm{Var}\left(U_0(\boldsymbol{\beta})\right) \end{pmatrix}$$

and verify that

$$\mathrm{Var}\left(\boldsymbol{U}(\boldsymbol{\beta})\right) \;=\; \boldsymbol{J}(\boldsymbol{\beta}) \;=\; \boldsymbol{I}(\boldsymbol{\beta})$$

where $\boldsymbol{J}$ and $\boldsymbol{I}$ are the observed and expected matrices of negative second derivatives of the log-likelihood, respectively.

Q. 3 [30 marks]

The `Fruitfly.csv` dataset (also available as `fruitfly` in the `faraway` package) contains data on the longevity of 125 male fruitflies and their thorax length. The flies were divided into 5 groups of 25. In the first group the flies were each kept individually, in the second group they were kept with one pregnant female per day, in the third each male was kept with 8 pregnant females per day, in the fourth group each male was housed with a virgin female per day and in the fifth group each male was housed with 8 virgin females per day. The group is in the `activity` column.

a) Make a plot of the longevity against thorax length differentiating between the 5 groups and comment on what the plot shows.

b) Fit a normal linear model with thorax length, activity and their interaction as covariates. Is there any evidence that the interaction is required? If not remove it from the model. Interpret the resulting fitted model.

c) Construct appropriate diagnostics and comment on any violations of the underlying assumptions that may be present.

d) Consider taking the log of the response variable and refit the model on the transformed response. Interpret the coefficients and repeat the diagnostic plots. What issues have been improved by this transformation and which remain possible problems?

e) Instead of a linear model use `glm` with `family=Gamma(log)` to fit a gamma generalized linear model with a log link and compare the estimated coefficients and their standard errors for the linear model on the log scale and for the generalized linear model.

f) Compare the fitted values from the gamma model in (e) with the models two linear models (raw response and log response).